

PHASE 5 – PROJECT DEVELOPMENT

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Development Overview

The Script-Controlled ACL project was developed using the ServiceNow platform. The main purpose of development is to control access to student records based on the user's role and the branch value of the record.

The development process includes creating users and roles, creating the Student Records table, adding branch-based records, configuring ACLs, and implementing scripts to control access.

2. User and Role Creation

The first step in development is creating the required users and roles in ServiceNow.

Users are assigned appropriate roles so that different access conditions can be tested.

The roles are used by the ACL scripts to identify whether a user is authorized to access a particular record.

3. Student Records Table Creation

A Student Records table is created in ServiceNow to store the records used in the project.

The table contains student information along with the branch value.

The branch field is important because the ACL script uses the branch value to determine whether access should be allowed or denied.

4. Creating Branch-Based Records

Student records are created with different branch values.

The project uses the following branch values:

ECE	EEE	CSE
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These records are used to verify whether users can access only the records for which they have permission.

5. Read ACL Development

A Read ACL is created to control the viewing of student records.

The ACL checks the user's role and the branch value of the record.

If the required conditions are satisfied, the user is allowed to view the record.

If the conditions are not satisfied, access to the record is restricted.

6. Write ACL Development

A Write ACL is created to control modification of student records.

The ACL checks whether the user has the required permission before allowing changes to the record.

This prevents unauthorized users from modifying restricted student records.

7. Create ACL Development

A Create ACL is implemented to control the creation of new student records.

The ACL checks the user's access permission before allowing a new record to be created.

Users who do not satisfy the required conditions are prevented from creating records.

8. Delete ACL Development

A Delete ACL is created to control the deletion of student records.

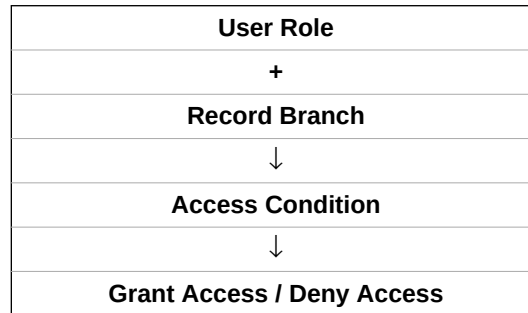
The ACL verifies the user's permission before allowing the delete operation.

This helps prevent unauthorized users from deleting protected records.

9. Script-Controlled Access Logic

The main feature of the project is the use of scripting inside ACLs.

The script checks:



The script-controlled conditions are applied to the required ACL operations. This allows access to be controlled according to the role of the user and the branch value of the record.

10. Administrator Access

Administrator access is configured separately.

Administrators are provided with full access to the Student Records table.

Therefore, administrators can perform the required operations on the records without the branch-based restriction applied to normal users.

11. Development Verification

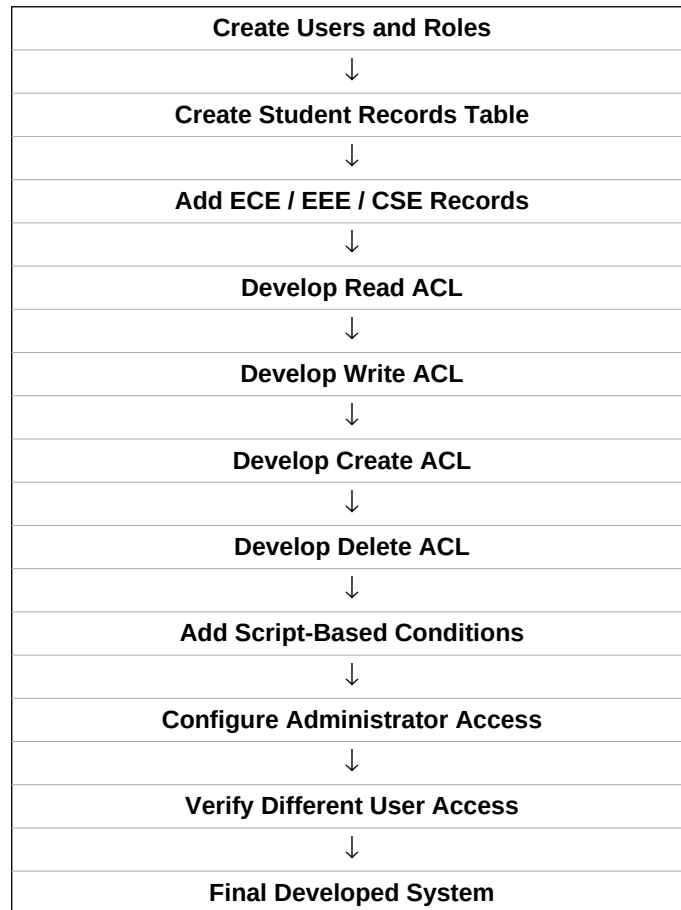
After implementing the ACLs and scripts, the configuration is checked using different users and roles.

The following operations are verified:

- Viewing student records.
- Modifying student records.
- Creating student records.
- Deleting student records.
- Accessing records belonging to different branches.

The results are used to confirm that the ACL configuration is working according to the project requirements.

12. Development Workflow



13. Development Outcome

The development process results in a ServiceNow-based access control system that uses Script-Controlled ACLs to protect student records.

The developed system provides:

- Role-based access control.
- Branch-based record restriction.
- Read access control.
- Write access control.
- Create access control.
- Delete access control.
- Full administrator access.
- Prevention of unauthorized record access.

14. Conclusion

The Script-Controlled ACL project was developed by configuring users, roles, student records, and ACLs in ServiceNow. Script-based conditions are used to verify user permissions and branch values before allowing access to records. The completed development provides controlled and secure access to the Student Records table.